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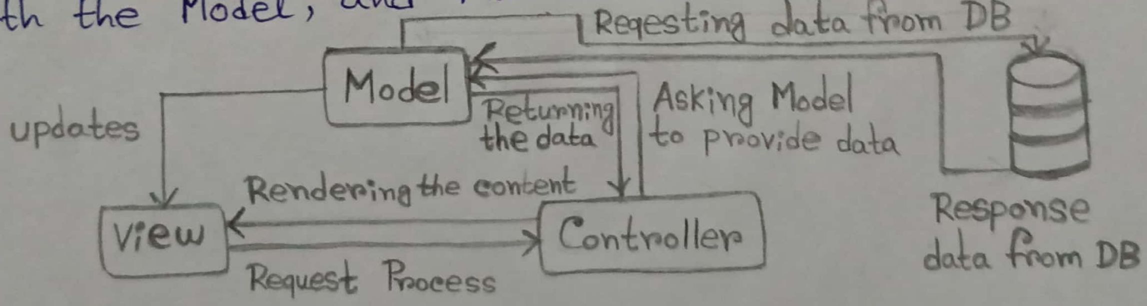
## Computer Architecture - Software
## Assignment - 1

### MVC Design Pattern:

The Model-View-Controller (MVC) pattern is a popular architecture style that divides applications into three interconnected components. This separation provides modular generation, faster tests and effective maintenance.

### Core Components of MVC:

- Model: Represents the data and business logic. It defines how data is manipulated and retrieved, often interacting with a database or external API.
- View: The presentation layer that shows data to customers. It collects and displays data from the Model, often using HTML, UI frameworks, or templates.
- Controller: The Controller serves as an intermediary between the view and Model. It accepts user input, communicates with the Model, and modifies the view appropriately.

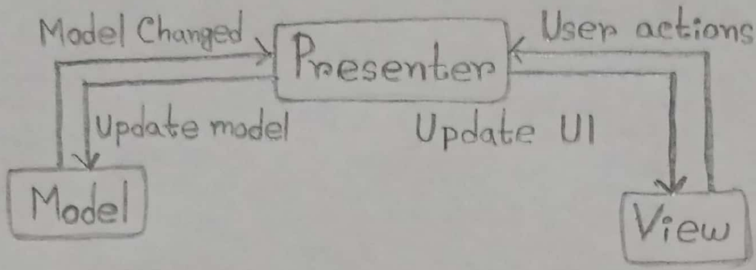



Dig. 1: MVC Architecture

### Two common variants of MVC:

a. Model-View-Presenter (MVP): The presenter replaces the Controller and handles both the View and the Model.

Unlike the Controller, the Presenter has a reference to the View and may update it directly.

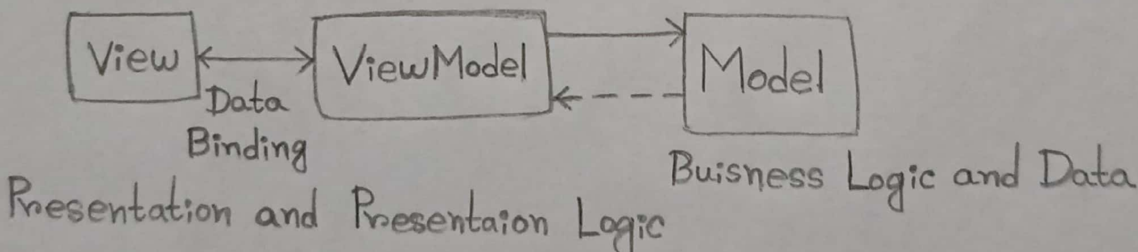



Dig.2: MVP Architecture

Use Case: MVP fits well in applications with complex user interactions, especially desktop or Android apps, where the view needs to be updated frequently and predictably.

b. Model-View-ViewModel (MVVM):

- Introduced by Microsoft, MVVM is popular in data-binding-friendly environments (such as WPF and Angular).
- The ViewModel exposes data from the Model via observables allowing the View to update automatically.




Dig.3: MVVM Architecture

Use Case: MVVM is ideal and best suited for UI-heavy, event-driven programms like SPA (Single Page Applications) or mobile apps with binding frameworks.

Choosing the appropriate alternative relies on the technology stack, UI complexity, and developer preferences. MVC provides a clear separation that is ideal for web frame-works, MVP gives greater control in user-intensive applications, and MVVM performs best when seamless data coupling is required.

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